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Machine Learning Applications in Employee Turnover and Performance Prediction: A Systematic Literature Review

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ABSTRACT

Organizations are increasingly adopting data-driven methods in human capital management, particularly for predicting employee performance and turnover, due to the growing availability of workforce data. Although machine learning has shown promise in this field, current research often examines performance and turnover prediction independently. However, employee performance plays a critical role in predicting turnover, as these two outcomes are interconnected—performance frequently influences individual decisions to remain with or leave a company. This study aims to identify frequently used models, input features, and evaluation metrics in machine learning, as well as to evaluate models that perform well on both prediction tasks. In accordance with PRISMA principles, a systematic literature review (SLR) of peer-reviewed research published between 2020 and 2025 was conducted. After applying predetermined inclusion and exclusion criteria, 23 papers were selected from academic databases (Scopus and Google Scholar). The review results demonstrate that tree-based ensemble models and neural network-based methodologies frequently outperformed other machine learning methods in predicting both performance and turnover. Models such as Random Forest, Gradient Boosting, XGBoost, and deep learning architectures (DNN, RNN-LSTM) delivered exceptional prediction accuracy, while hybrid models further enhanced outcome explainability, reliability, and practical application in HR analytics. These findings offer critical managerial implications by guiding HR practitioners in selecting appropriate predictive models for workforce planning, retention strategies, and performance management systems. They also highlight key research gaps, including dataset bias, ethical considerations in algorithmic decision-making, and the need for longitudinal validation studies across diverse organizational contexts.

INTRODUCTION

Machine learning (ML) is a field within artificial intelligence that enables computers to learn from past information, recognize patterns, and make judgments with minimal human input to predict future outcomes. ML uses a variety of methods, such as ensemble approaches, decision trees, regression, and support vector machines (Bzdok, Krzywinski, & Altman, 2018).

In the current competitive business environment, firms continually seek methods to enhance their efficiency and performance. Predicting staff performance and productivity is crucial for organizational success. High-performing employees play a critical role in achieving organizational objectives, and their retention represents a key source of competitive advantage in the business environment (Poddar & Chattopadhyay, 2021). One promising tool to support this objective is machine learning (ML), which can substantially enhance human resource management by optimizing employee performance and turnover prediction processes, making them more efficient, effective, and objective (Indarapu et al., 2023).

Employee turnover represents a critical challenge for organizations, as it is unpredictable and usually results in significant gaps in the skilled workforce. Recent advancements in machine learning have provided powerful tools for analyzing large and complex datasets (Ali Shah et al., 2020;Govindarajan et al., 2025). Supervised machine learning methods, which learn from historical labeled data, have been successfully applied in many fields. With innovations in information technology, these methods are increasingly used to support and improve human resource management practices, particularly in analyzing employee turnover (Zhao et al., 2018).

Historically, assessments of employee turnover and performance were made based on a limited amount of data and subjective evaluations (Atef et al., 2022; Deviprasad et al., 2023). As a result, these decisions have always been less than ideal. Machine learning models powered by artificial intelligence come to the rescue by utilizing extensive quantities of data and advanced analytical skills (Rathore & Rathore, 2024).

Recent evidence from machine learning research indicates that employee performance plays a critical role in predicting turnover. Bernard et al. (2025) demonstrate that turnover prediction models achieve substantially higher accuracy when performance data are included, compared to models that exclude such information. This finding emphasizes the importance of employee performance as a key predictive feature in machine learning approaches to employee turnover analysis. Existing meta-analytic studies have also shown a clear and consistent relationship between job performance and employee turnover. In addition, prior research has developed and tested process-based models that explain how job performance influences employees' decisions to leave an organization (Diamantidis & Chatzoglou, 2019; Goodell et al., 2021). These studies suggest that the link between performance and turnover is not merely correlational but operates through identifiable mechanisms that shape employee behavior. According to a meta-analysis of 24 studies, high-performing employees are less likely to leave their companies than low-performing ones. This supports negative performance–turnover models rather than those that indicate high performance enhances turnover through ease of transferring jobs (Alzubi et al., 2018; Halder et al., 2024).

Despite the substantial body of literature on machine learning applications in HR analytics, a critical research gap persists: the majority of existing studies examine employee performance prediction and turnover prediction as independent, isolated tasks, failing to acknowledge the empirically established interdependence between these two outcomes. While meta-analytic evidence confirms that performance significantly influences turnover decisions, and recent studies demonstrate improved turnover prediction accuracy when performance data are incorporated (Bernard et al., 2025), no comprehensive systematic review has synthesized the methodological landscape across both prediction domains simultaneously. This gap is particularly problematic given that organizations require integrated predictive frameworks that can assess both current performance levels and future attrition risk to inform holistic workforce management strategies.

The novelty of this study lies in three key contributions. First, it provides the first systematic literature review that explicitly examines machine learning applications across both employee performance and turnover prediction tasks, identifying algorithms, features, and metrics that demonstrate robust performance in both domains. Second, it synthesizes findings from 23 peer-reviewed studies published between 2020 and 2025, capturing the most recent methodological innovations, including explainable AI, hybrid models, and time-to-event approaches that represent the current state-of-the-art in HR analytics. Third, it offers practical guidance for HR practitioners by identifying high-performing models suitable for dual prediction tasks, thereby enabling organizations to implement integrated predictive systems that simultaneously monitor performance and anticipate attrition risk, leading to more proactive and strategic human resource management.

The findings of this systematic review offer several critical practical implications for organizational stakeholders. For HR practitioners and managers, this study provides evidence-based guidance on selecting appropriate machine learning algorithms for workforce analytics initiatives, identifying tree-based ensemble models (Random Forest, XGBoost, LightGBM) and deep learning architectures (DNN, RNN-LSTM) as optimal choices for achieving high predictive accuracy across both performance evaluation and turnover forecasting. Organizations can leverage these insights to design integrated HR analytics systems that simultaneously monitor employee performance trajectories and identify early warning signals of potential attrition, enabling proactive interventions such as targeted retention programs, personalized development plans, and strategic succession planning.

For technology vendors and HR software developers, the review highlights the growing demand for explainable AI capabilities (SHAP, LIME) that enhance model transparency and trust, suggesting that future HR analytics platforms should prioritize interpretability alongside predictive power to facilitate managerial decision-making and ensure ethical compliance. Additionally, the identification of critical input features across studies provides organizations with a blueprint for data collection priorities, emphasizing the importance of capturing behavioral indicators (overtime, absenteeism), engagement metrics (satisfaction, job involvement), and longitudinal performance records to maximize model effectiveness. Finally, for organizational leaders and policymakers, the review underscores the strategic value of investing in data infrastructure and analytical capabilities as foundational elements of competitive advantage in talent management, while also highlighting ethical considerations regarding algorithmic fairness, privacy protection, and the responsible use of predictive analytics in employment contexts.

METHOD

The study includes articles published between 2020 and 2025 from Scopus and Google Scholar since this time period represents a substantial shift in both the development of machine learning techniques and their practical implementation in HR analytics. Beginning in 2020, advances in computational power, the availability of large-scale HR datasets, and the widespread adoption of advanced machine learning models such as ensemble methods, deep learning, explainable AI, and hybrid approaches have significantly improved the predictive accuracy and interpretability of employee performance and turnover models. Google Scholar was used as another source to capture potentially relevant research not indexed by Scopus, particularly recent research papers from conferences and interdisciplinary journals. All Google Scholar entries were cross-checked against Scopus results to avoid duplication.

For the Scopus database, a structured Boolean search strategy was employed by combining machine learning related terms with employee turnover and performance. Specifically, the search string included variations of machine learning in conjunction with employee turnover terms. Keyword 1, such as (“machine learning algorithm" OR "machine learning") AND ("employee retention" OR "employee turnover prediction" OR "turnover" OR "employee attrition" OR "attrition" OR "employee churn" OR "churn") as well as performance terms. Keyword 2, such as ("machine learning" OR "machine learning algorithm") AND ("employee performance" OR "performance prediction") also ("employee performance" AND "machine learning").

In addition, Google Scholar was used as a complementary source, applying simplified keyword combinations “machine learning employee performance” and “machine learning employee turnover”. This study performed a systematic review following the best practice of PRISMA methods as shown in Figure 1 (Mishra & Mishra, 2023).

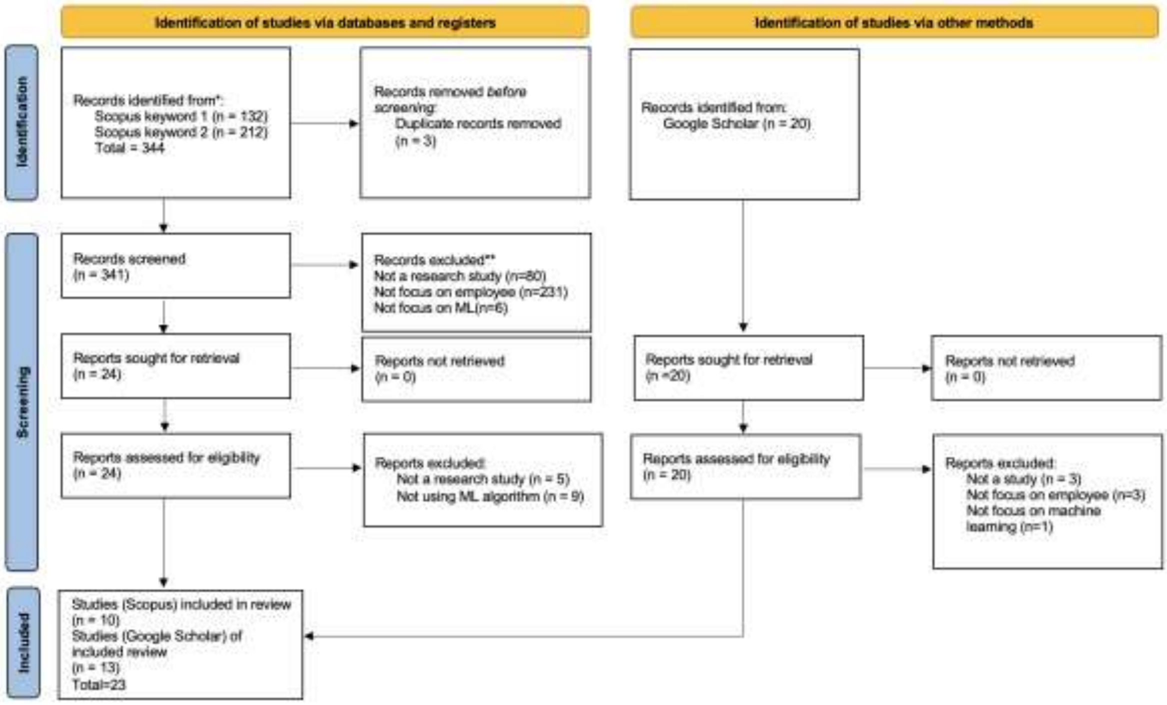

Figure 1. PRISMA Diagram

This systematic literature review adhered to PRISMA guidelines to maintain transparency and reproducibility in the study selection process. The initial search yielded 344 records from Scopus, using two keyword strategies, which were narrowed to 341 unique records after removing 3 duplicates. An additional 20 records were sourced from Google Scholar through complementary search terms.

During screening, 341 Scopus records were examined, with 317 excluded for various reasons, such as not a research studies (80), not focused on employees (231), or not involving machine learning (6). This left 24 Scopus records for full-text retrieval, all of which were successfully obtained. The 20 records from Google Scholar were also screened and retrieved without exclusions.

Eligibility assessment of the 24 Scopus articles resulted in the exclusion of 14 articles due to reasons including lack of research methodology (5) or absence of machine learning algorithm (9), leading to 10 eligible studies. From the 20 Google Scholar articles, 7 were excluded for lack of relevance to the employee context (3), absence of machine learning focus (1), or not a research studies (3), resulting in 13 eligible studies.

Ultimately, 23 studies were included in the final review, which are 10 from Scopus and 13 from Google Scholar, forming the basis for the qualitative synthesis on machine learning approaches for predicting employee performance and turnover.

RESULTS AND DISCUSSION

Machine Learning Used in Predicting Employee Performance and Turnover

1. Machine Learning Approaches Used in Predicting Employee Turnover

The majority of the research evaluated focus on predicting employee turnover, with the problem often presented as a binary classification job that distinguishes between individuals who stay and those who leave an organization. Throughout the study shows in Table 1, supervised machine learning models are continuously used, with a focus on tree-based and ensemble learning techniques.

Table 1. Machine Learning Used in Employee Turnover Prediction

No	Title	Year	Analytical Model	Machine Learning Algorithm	Comparison
1	Intelligent Employee Retention System for Attrition Rate Analysis and Churn Prediction: An ensemble machine learning and multi-criteria decision-making approach	2021	a regression model and a multi-criteria fuzzy analytical hierarchy process (AHP)	Deep neural network, Random Forest, gradient boosting	Compared each ML Algorithm
2	Enhancing employee churn prediction with weibull time-to-event modeling	2025	Weibull Time-to-Event - Recurrent Neural Network forecasting model (WTTE-RNN)	Recurrent Neural Network (RNN)	Traditional machine learning model (Logistic regression, random forest, XGBoost, ARIMA, SARIMAX, LSTM)
3	Predicting Employee Attrition: A Comparative Analysis of Machine Learning Models Using the IBM Human Resource Analytics Dataset	2025	N/A	Random forest, Support Vector Machine (SVM), Decision tree, Gradient Boosting, Hybrid model	Compared each ML Algorithm
4	An integrated model to evaluate the transparency in predicting employee churn using explainable artificial intelligence	2025	Shapley Additive exPlanations approach (SHAP)	Logistic Regression (LR), Support Vector Machine (SVM), K-Nearest Neighbors (KNN), Random Forest (RF), Decision Tree Classifier, and Gaussian Naive Bayes (GNB)	Compared each ML, the highest Algorithm further evaluation using SHAP

No	Title	Year	Analytical Model	Machine Learning Algorithm	Comparison
				were applied using Python	
5	Predicting employee attrition and explaining its determinants	2025	Shapley Additive exPlanations approach (SHAP)	Naive bayes, logistic regression, decision tree, random forest	Compared each ML, the highest Algorithm further evaluation using SHAP
6	Developing a hybrid machine learning model for employee turnover prediction: Integrating LightGBM and genetic algorithms	2025	GA + LightGBM model	LightGBM and genetic algorithms	Conventional machine learning models: Random Forest, Logistic Regression, K-Nearest Neighbours, Support Vector Machine, Decision Tree, XGBoost, Naive Bayes, and LightGBM (standalone)
7	Application of data visualization technology in human resource management and employee resignation prediction	2025	Improved CART-DT (Classification and Regression Tree - Decision Tree) and cross-validation method (CVM)	Improved CART-DT	Naive Bayes, logistic regression, and gradient boosting tree
8	Optimization-based Techniques Prediction Model in Determining Employee Turnover	2025	4 crossover (AX, CAX, IBAX, and FMSAX)	GA (Genetic Algorithms) + Naive Bayes and KNN	Compared each ML Algorithm
9	A cluster-based human resources analytics for predicting employee turnover using optimized Artificial Neural Networks and data augmentation	2024	Artificial Neural Network (ANN) and data augmentation CTGAN (Conditional Generative Adversarial Networks)	Artificial Neural Network (ANN) using cluster	Optimized ANN (without augmentation, Neural Network, and four traditional machine learning models (K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Naive Bayes (NB), Logistic Regression (LR))
10	Explainable Machine Learning and Graph Neural Network Approaches for Predicting Employee Attrition	2024	Explainable GNN (interpreting the factor of turnover)	Graph neural networks (GNNs)	Traditional machine learning models achieve high accuracy (e.g., XGBoost: 95.55% accuracy, 98.44% recall)
11	Early Prediction of Employee Turnover Using Machine Learning Algorithms	2022	N/A	KNN (K-Nearest Neighbors) and Random Forest (RF)	Compared each ML Algorithm

No	Title	Year	Analytical Model	Machine Learning Algorithm	Comparison
12	Developing an advanced prediction model for new employee turnover intention utilizing machine learning techniques	2024	OLS regression analysis	Logistic Regression (LR), k-Nearest Neighbor (KNN), Extreme Gradient Boosting (XGB) classifier	Compared each ML Algorithm
13	Leveraging Machine Learning Methods for Predicting Employee Turnover Within the Framework of Human Resources Analytics	2024	R-Programming Language	Decision trees, support vector machines (SVM), logistic regression, random forest, XGBoost, Artificial Neural Networks (ANN)	Compared each ML Algorithm
14	Churn Prediction of Employees Using Machine Learning Techniques	2021	N/A	Random Forest Classifier, SVM (Support Vector Machine), Naive Bayes Classifier	Compared each ML Algorithm
15	Using Human Resources Data to Predict Turnover of Community Mental Health Employees: Prediction and Interpretation of Machine Learning Methods	2024	Cross-Validation Method (CVM) and Shapley Additive exPlanations approach (SHAP)	Logistic Regression (LR), Elastic Net (EN), Random Forest (RF), Gradient Boosting Machine (GBM), Neural Network (NN), Support Vector Machine (SVM)	Compared each ML Algorithm
16	A Predictive Approach and Recommendation System for Employee Turnover Using Machine Learning Algorithms	2025	CRISP methodology and Local Interpretable Model Agnostic Explanations (LIME)	Random forest, Logistic Regression, and Support Vector Machines	Compared each ML Algorithm

In addition to ensemble approaches, numerous research uses traditional classification algorithms like Logistic Regression, Decision Tree, Naive Bayes, K-Nearest Neighbors, and Support Vector Machines to serve as baseline models for performance comparison. While these models have lower prediction accuracy than ensemble techniques, they are still helpful because of their interpretability and ease of implementation. Random Forest and Gradient Boosting models, including variants such as XGBoost and LightGBM are preferred algorithms because they can capture non-linear correlations, manage high-dimensional HR data, and maintain relatively robust predictive performance across various business contexts. Neural network-based techniques, such as Artificial Neural Networks and Deep Neural Networks, are also becoming more common in turnover prediction, particularly in studies aimed at improving predictive accuracy through advanced, multi-layered designs. Multi-layer neural networks process employee data through several hierarchical stages, allowing the model to learn increasingly complex representations related to employee turnover risk.

More recent studies use time modelling approaches to handle the dynamic nature of employee turnover. Models like the Weibull Time-to-Event Recurrent Neural Network (WTTE-RNN) extend beyond traditional categorization frameworks by predicting not just the likelihood of turnover but also the timing of employee leaving. These methods frequently outperform traditional machine learning models when longitudinal or sequential employee data is available, indicating a methodological change towards more realistic and time-sensitive turnover prediction.

Based on reviewed studies, machine learning models for employee turnover prediction are often combined with additional analytical methods to make the results more reliable, understandable, and useful for real human resources management decisions. Some research also uses explainable artificial intelligence techniques, such as SHAP interpretations, to improve the transparency of complex machine learning models and assist with managerial decision-making. Other methods, such as the Analytical Hierarchy Process and CART decision trees, are used to help decision-makers understand which factors matter most and why a prediction is made, rather than relying on black box results. Cross-validation techniques are applied to ensure that the model's performance is stable and not dependent on one specific data split, which is especially important when HR data are limited or imbalanced. Frameworks like CRISP (Cross-Industry Standard Process) are adopted to guide the entire analysis process step by step, ensuring that the modeling aligns with actual organizational goals. More advanced methods, such as CTGAN (Conditional Generative Adversarial Networks) and explainable AI techniques like LIME (Local Interpretable Model Agnostic Explanations), are used to handle data imbalance, protect sensitive employee information, and explain individual predictions in simple terms.

2. Machine Learning Approaches Used in Predicting Employee Performance

Employee performance prediction, unlike turnover prediction, has received relatively limited attention and remains a less explored research area in the application of machine learning. The machine learning studies on employee performance prediction presented in Table 2 primarily frame performance prediction as either a classification problem, such as identifying high and low performing employees, or as a regression task aimed at predicting performance scores or assessment outcomes. Similar to turnover research, the reviewed studies employ a wide range of supervised machine learning algorithms, including regression models, tree-based methods, distance-based classifiers, probabilistic approaches, and neural network architectures. Ensemble techniques such as Random Forest, Gradient Boosting, AdaBoost, and Bagging are frequently used to enhance predictive performance. In addition, several studies apply data preprocessing and methodological techniques, including PCA for dimensionality reduction, ROS and SMOTE for class imbalance handling, and CRISP-DM as a data mining framework.

Table 2. Machine Learning Used in Employee Performance Prediction

No	Title	Year	Analytical Model	Machine Learning Algorithm	Comparison
1	Predicting Accurate Employee Performance: An Evaluation of Regression Models	2025	Regression techniques	Linear Regression, Decision Branch/Tree model, Random Decision Forest, Gradient Boosting classifier, KNN, SVM, and Neural Network	Compared each ML Algorithm

No	Title	Year	Analytical Model	Machine Learning Algorithm	Comparison
2	Employee Promotion Prediction by using Machine Learning Algorithms for Imbalanced Dataset	2022	Imbalanced technique method: ROS (Random OverSampling) and SMOTE (Synthetic Minority Over-Sampling Technique)	Support Vector Machine (SVM), Artificial Neural Network (ANN) dan Random Forest (RF)	Compared each ML Algorithm using data from ROS and SMOTE techniques
3	Comparative Analysis of Machine Learning Techniques for the Prediction of Employee Performance	2022	Principal Component Analysis (PCA)	Artificial Neural Network (ANN), Random Forest (RF), Decision Tree (DT)	Compared each ML Algorithm
4	Towards a new method for classifying employee performance using machine learning algorithms	2022	CRoss Industry Standard Process for Data Mining (CRISP-DM) and Principal Component Analysis (PCA)	K-Nearest Neighbors (K-NN)	Previous Study with the factor identified: Workforce Diversity, Training Determinants of employee engagement, employee satisfaction, and Leadership Style
5	Real-Time Employee Performance Monitoring and Prediction Using RNN-LSTM with Attention Mechanism for HR Analytics	2025	Principal Component Analysis (PCA)	RNN-LSTM model with an Attention Mechanism	RNN-LSTM compared with SVM Method and Decision Tree Method
6	An enhanced deep neural network for predicting workplace absenteeism	2020	N/A	Deep neural network	Deep neural network compared to single-layer neural network (shallow neural network), decision tree, SVM (support vector machine), random forest
7	A Combined Approach for Predicting Employees' Productivity based on Ensemble Machine Learning Methods	2022	Ensemble Techniques (AdaBoost and Bagging)	J48, Random Forest, Radial Base Function Network (RBF), Multilayer Perceptron (MLP), Naive Bayes (NB), and Support Vector Machine (SVM)	Ensemble Algorithms vs. Standard Algorithms

Other algorithms for predicting employee performance are Naive Bayes and Support Vector Machines. The application of deep learning models, including Artificial Neural Networks (ANNs), Deep Neural Networks (DNNs), and Recurrent Neural Networks (RNNs), are applied to employee performance prediction because they are designed for modeling the complex, non-linear, and dynamic nature of human performance data. Employee performance is rarely influenced by a single factor. It emerges from interactions among multiple variables such as skills, workload, training history, engagement, collaboration, and organizational context. Deep learning models can automatically learn these complex relationships.

Similarities and Differences in Models, Features, and Evaluation Metrics

Employee Turnover Prediction

Across the reviewed papers, the input features used to train machine learning models for employee retention or turnover prediction vary significantly, as presented in Table 3. The studies reflect variances in data availability, organizational context, and research objectives. Despite this variety, most studies follow a common conceptual structure of employee turnover factors, which includes individual, job specific factor, organizational, and performance dimensions.

Commonly utilized characteristics include age, gender, status, education, tenure, job position, salary or compensation level, working hours, job satisfaction, performance ratings, promotion history, and training participation. Several studies additionally include behavioral and engagement variables, including absenteeism, overtime, and workload, while more recent research adds longitudinal performance records and temporal job histories. The addition of employee performance factors is also important since several studies have shown that performance indicators significantly enhance turnover prediction accuracy.

Table 3. Input Features, Accuracy, and Evaluation Metrics Turnover Prediction

No	Title	Input features	Accuracy	Evaluation metrics
1	Intelligent Employee Retention System for Attrition Rate Analysis and Churn Prediction: An ensemble machine learning and multi-criteria decision-making approach	Employee satisfaction, appraisal rating, employee CTC level, number of projects/tasks assigned per quarter, time spent per project per quarter, promotion, safety measure	**Deep neural network (91,6%)**, Random forest (82,5%), gradient boosting (85,4%)	Accuracy, Root Mean Square Error
2	Enhancing employee churn prediction with weibull time-to-event modeling	joining year, payment tier, age, satisfaction level, last evaluation, experience in current domain, leave or not, number project, average monthly hours, time spent company, work accident, left, promotion last 5 year	N/A	Mean Squared Error (MSE) and R-Squared (R2)
3	Predicting Employee Attrition: A Comparative Analysis of Machine Learning Models Using the IBM Human Resource Analytics Dataset	Age, daily rate, distance from home, education, environment satisfaction, hourly rate, job involvement, job level, job satisfaction, mothly income, monthly rate, number companies worked, percent salary, performance rating, relationship satisfaction, stock option level, total working years, training times last year, work life balance, years at company, years in current role, years since last promotion, years with current manager.	Random forest and gradient boosting (87,3), support ventor machine (SVM) (88,6%), Decision tree (80,5%)**, Hybrid model (95%)**	Accuracy, Confusion Matrix

No	Title	Input features	Accuracy	Evaluation metrics
4	An integrated model to evaluate the transparency in predicting employee churn using explainable artificial intelligence	Overtime, monthly income, stock option level, job level, age, total working years, distance from home, marital status, years at company, number companies worked, environment satisfaction, years with current manager, years in current role, job satisfaction, department sales, business travel frequently, daily rate, years since last promotion, department, job role, education field, relationship satisfaction, job involvement	Logistic regression (87,96%) - (overtime and marital status) and random forest analysis (overtime and monthly income) (87,29%)	Accuracy, Precision, Recall, F1- score and AUC
5	Predicting employee attrition and explaining its determinants	Bonus, retention, hours, talent, rank, department, contract, gender, level, age class, tenure class, job description, log salary, log premium	Naive bayes (74,8-75,6%), logistic regression (68,8-78,9%), decision tree (76,3-81,7%), random forest (94,7% - 95,9%)	Accuracy, Sensitifity, Specificity, AUC
6	Developing a hybrid machine learning model for employee turnover prediction: Integrating LightGBM and genetic algorithms	Age, monthly income/salary, job satisfaction, work life balance, environment satisfaction, job involvement, education level, performance rating, years at company, years in current role, years since last promotion, total working years, distance from home, stock option level, number companies worked, gender, overtime, job role, department, daily rate, hourly rate, monthly rate, training times last year, standard hours, employee number, employee count	GA+LightGBM (92%) , Logistic Regression (75%), Random Forest (78%), K-Nearest Neighbour (74%), Support Vector Machine (72%), Decision Tree (76%), Naive Bayes (77%), XGBoost (81%), LightGBM (86%)	Accuracy, Precision, Recall, F1-Score, and AUC
7	Application of data visualization technology in human resource management and employee resignation prediction	employee gender, job position, monthly salary, length of service, working hours, average performance evaluation, job satisfaction, commuting distance, whether promoted, whether major work errors have occurred, and employee educational level	Improved CART-DT (96,54%), logistic regression (91,04%), gradient boosting decision tree (GBDT) (89,62%), Naive Bayer (81,65%)	Accuracy, Error Rate, Precision, Recall, F1-Score, and AUC
8	Optimization-based Techniques Prediction Model in Determining Employee Turnover	Health problem, inadequate compensation, family-related obligations or priorities, pursue study, difficult transportation, workload & stress, work–life balance, co-workers, contractual, looking for higher salary, conflict in management, unfair pay, a desire to start their own business, relocation, inflexible hours, retirement or career transition, benefits, bad boss, pursue study, company culture, company financial instability, feeling undervalued, no personal vehicle, inadequate compensation.	Initial Population: KNN (73,32%), Naive Bayes (46,45%) The Highest Crossover: FMSAX + KNN (95,52%)	Accuracy and F-measure
9	A cluster-based human resources	satisfaction, last evaluation, number project, average monthly hours, time spend	Cluster: ANN with augmentation (91,66%-	Accuracy, Recall

No	Title	Input features	Accuracy	Evaluation metrics
	analytics for predicting employee turnover using optimized Artificial Neural Networks and data augmentation	company, work accident, employee left, promotion last 5 years, department, salary	93,83%), NN 1 (87,79 - 95,23%), NN 2: (89,15%-96,91%) NN3: (92,78%-95,63%), NN4: (87,48%-95,69%), NN5 : (88,9%-92,60%), KNN: (85,7%-94,8%), SVM: (73,3%-92%), NB: (31,6%-87,2%), LR: (79,1%-92,1%) Non Cluster: NN (94,9%), KNN (93,87%), LR (75,8%), SVM (79,47%), Naive Bayes (79,73%)	
10	Explainable Machine Learning and Graph Neural Network Approaches for Predicting Employee Attrition	Age, Attrition, BusinessTravel, DailyRate, Department, DistanceFromHome, Education, EducationField, EmployeeNumber, EnvironmentSatisfaction, Gender, HourlyRate, JobInvolvement, JobLevel, JobRole, JobSatisfaction, MaritalStatus, MonthlyIncome, MonthlyRate, NumCompaniesWorked, OverTime, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance,YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager.	Graph Neural Network (70,4%), Recurrent Neural Network (63,2%), Graph Attention Network (74,3%), Feedforward Neural Network (57,5%), Artificial Neural Network (89,5%), Adaptive Gradient Boosting Classifier (95,6%), Stochastic Gradient Descent Classifier (72,3%), Logistic Regression (76,3%), Adaptive Boosting (78,5%), Support Vector Machine (86%), Random Forest (85%), K-Nearest Neighbors (82%), Decision Trees (90%)	Accuracy, Precision, Recall and F1-Score
11	Early Prediction of Employee Turnover Using Machine Learning Algorithms	Salary (monthly income), age, distance from home, marital status, gender	KNN (83,5%), RF (78,4%)	Accuracy, Precision, F-score, Specificity (SP), False-Positive Rate (FPR)
12	Developing an advanced prediction model for new employee turnover intention utilizing machine learning techniques	Extrinsic motivation (job preference, existence needs, relatedness needs), intrinsic motivation (growth needs) and personal-job fit.	Cross Validation Average LR (78,3%), KNN (76,1%), XGB (78,5%)	Accuracy, precision, recall, and F1-score
13	Leveraging Machine Learning Methods for Predicting Employee Turnover Within the Framework of	Demographic Independent Variable : Age, Marital Status, Gender, Education, Travel Status Job-related Independent Variable: Daily Wage, Department, Commute Distance, Field of Study, Environmental	LR (89,8%), ANN (88,78%), RF (86,05%), XGBoost (87,07%), Decision Tree (86,05%), SVM (85,03%)	Accuracy, Sensitivity, Specificity, Precision, F1-Score, ROC-AUC

No	Title	Input features	Accuracy	Evaluation metrics
	Human Resources Analytics	Satisfaction, Engagement Level, Work-Family, Role, Job Satisfaction, Monthly Income, Salary Raise, Number of Companies Worked At, Job Satisfaction, Overtime, Salary Raise %, Performance Rating, Communication Satisfaction, Working Hours, Stock Option Level, Work Experience, Training Duration (Last Year), Work-life Balance, Seniority, Tenure in Role, Years with Current Manager Dependent Variable: Attrition Status		
14	Churn Prediction of Employees Using Machine Learning Techniques	Age, Gender, Salary Level, Years of Experience, Satisfaction Level, Discrimination, Work Recognition, Challenging Work, Promotion in Last Year, Peers Leaving, Quit in 2 years.	Case 1 (Comparing the performance metrics for all the Target Variable classes): Random Forest (70,83%), Naive Bayes (68,75%), and SVM (60,42%) Case 2 (Comparing the performance metrics for only two Target Variable classes (without "Maybe"): Random Forest (77,42%), Naive Bayes (77,42%), and SVM (70,97%)	Confusion Matrix, Recall, False Positive Rate, and Accuracy
15	Using Human Resources Data to Predict Turnover of Community Mental Health Employees: Prediction and Interpretation of Machine Learning Methods	Demographics & Job Info (Past work years, average work hours, wage, age, exempt status, educational degree, marital status, and employee type) and mental health spesific predictors (training hours, proportion of clients with a schizophrenia diagnosis)	N/A	Specificity, Sensitivity, and AUC
16	A Predictive Approach and Recommendation System for Employee Turnover Using Machine Learning Algorithms	Department, Marital Status, Number of Children, Age, Salary, Category, Year of Employment, Number of Years of Service	Logistic regression (77%), Support vector machine (SVM) (82%) and Random Forest (97%)	Accuracy, Precision, Recall, F1-score, Confusion Metrics

In terms of model accuracy, employee turnover prediction using machine learning is typically high, especially in research that uses ensemble and deep learning techniques. Accuracy scores frequently approach 80%, with numerous studies finding values of over 90% when performance data or temporal factors are included. However, accuracy varies greatly based on data quality, feature selection, and class imbalance treatment. Studies using time-to-event models, such as recurrent neural networks with Weibull distributions, outperform traditional classification methods, particularly when predicting the timing of turnover rather than just the probability.

In terms of evaluation metrics, turnover prediction studies use a wide range of evaluation criteria. Accuracy is the most widely reported statistic, followed by precision, recall, F1-score, and AUC-ROC. Accuracy measures the overall proportion of correct predictions and is frequently reported due to its simplicity. However, in employee turnover prediction, accuracy can be misleading because turnover events are relatively rare, and several studies explicitly indicate that accuracy alone is insufficient due to the class imbalance between those who stay and those who leave. As a result, additional metrics such as precision, recall, F1-score, and AUC-ROC are necessary to provide a more comprehensive evaluation. Precision reflects the reliability of predicted turnover cases, recall captures the model's ability to identify actual leavers, the F1-score balances both measures, and AUC-ROC assesses the model's overall discriminative capability across different decision criteria.

Employee Performance Prediction

Employee performance prediction studies, on the other hand, rely on a more limited and structured collection of input characteristics, which are primarily concerned with individual productivity, assessment results, behavioral indicators, and task-related factors. Table 4 presents an input features, accuracy and evaluation metrics of machine learning models for employee performance studies. Common elements include performance ratings, key performance indicators (KPI), task completion rates, attendance records, and supervisor evaluations. Demographic factors are considered less often than in turnover studies, indicating a greater emphasis on job-specific and behavioral factors of performance.

Table 4. Input Features, Accuracy, and Evaluation Metrics Performance Prediction

No	Title	Input features	Accuracy	Evaluation metrics
1	Predicting Accurate Employee Performance: An Evaluation of Regression Models	Key features: Employee demographics, Job roles, Work hours, Past performance, Feedback scores Key determinant identified from ensemble model like random forest and gradient boosting classifier: job role, years of experience, and past performance scores	N/A	Mean Squared Error (MSE), R^2 score, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE).
2	Employee Promotion Prediction by using Machine Learning Algorithms for Imbalanced Dataset	Seniority, performance level, competencies, age, awards, training score, organizational commitment. Attribute: employeeid, department, region, education, gender, recruitment channel, no of training completed, age, previous year rating, length of service, awards won, avg training score, target promoted	SVM (92%), SVM_ROS (72,6%), SVM_SMOTE(75,6%), ANN (92,34%), ANN_ROS (72,9%), ANN_SMOTE (77%), RF (93,14%), RF_ROS (98,42%), RF_SMOTE (91,57%)	Accuracy, Precision, Recall, Sencifity, and F1-Score
3	Comparative Analysis of Machine Learning Techniques for the	Names, dates of birth, marital status, gender, date of hire, department, grounds for termination, active or terminated	ANN (98,94%), RF (97,87%), DT (96,81%)	Accuracy, Precision, Specificity,

No	Title	Input features	Accuracy	Evaluation metrics
	Prediction of Employee Performance	status, position title, salary rate, manager name, and performance score		Sensitivity, F1 Score
4	Towards a new method for classifying employee performance using machine learning algorithms	Employee Name, EmpID, MarriedID, MaritalStatusID, EmpStatusID, DeptID, PerfScoreID, FromDiversityJobFairID, PayRate, Termd, PositionID, Position, State, Zip, DOB, Sex, MaritalDesc, CitizenDesc, HispanicLatino, RaceDesc, DateofHire, DateofTermination, TermReason, EmploymentStatus, Department, ManagerName, ManagerID, RecruitmentSource, PerformanceScore, EngagementSurvey, EmpSatisfaction, SpecialProjectsCount, LastPerformanceReviewDate, DaysLateLast30	KNN using PCA (85,71%)	Accuracy
5	Real-Time Employee Performance Monitoring and Prediction Using RNN-LSTM with Attention Mechanism for HR Analytics	Task completion rate, work efficiency, absenteeism patterns, and level of engagement, demographic attributes (employee age, department, work hours)	RNN-LSTM with attention mechanism (99,5%), SVM Method (85,3%), Decision Tree (88,7%)	Accuracy, Precision, Recall, F1-Score, ROC-AUC
6	An enhanced deep neural network for predicting workplace absenteeism	Reason for absence, Month of absence, Day of the week, Seasons, Transportation expense, Distance from residence to work, Service time, Age, Work load average/day, Hit target, Disciplinary failure, Education, Socialdrinker, Social smoker, Pet, Weight, Height, Body mass index, and Absenteeism category	Deep Neural Network (DNN) (90,6%), Shallow Neural Network (Single-Layer NN) (73,3%), SVM (84,32%), Decision tree (82,83%), Random Forest (82,43%)	Accuracy, True Positive (TP) Rate, False Positive (FP) Rate, F1-score, ROC (Receiver Operating Characteristic Area), Confusion Matrix)
7	A Combined Approach for Predicting Employees Productivity based on Ensemble Machine Learning Methods	date of work, day of work, department, number of team, number of workers, number of style change of a particular product, targeted productivity, standard minute value, number of work in progress product, overtime, incentive, idle time, idle men, and actual productivity	Random forest single classifier (98,3%), MLP ensemble bagging (98,6%), J48 ensemble boosting (99,16%)	Accuracy, Mean Absolute Error (MAE), Root Mean Square Error (RMSE)

The accuracy findings from the investigated experiments demonstrate that ensemble-based and deep learning models frequently outperform others in predicting employee performance. Among traditional algorithms, Random Forest (RF) exhibits robust and consistent accuracy, especially when integrated with data balancing methodologies. The RF-ROS model achieves the best accuracy at 98.42%, surpassing ordinary RF and SMOTE-based variations, hence emphasizing the effectiveness of oversampling techniques in enhancing predictive performance. Similar to that, Artificial Neural Networks (ANN) provide competitive outcomes, with accuracy significantly improving with optimization, reaching 98.94%

Advanced deep learning methodologies significantly improve prediction precision. The RNN-LSTM model, including an attention mechanism, attains the greatest documented accuracy (99.5%), demonstrating its ability to identify sequential and temporal patterns in employee performance data. Deep Neural Networks (DNN) exceed shallow neural networks and traditional classifiers, indicating the advantages of multi-layer designs in capturing complex relationships.

Ensemble learning methods frequently provide resilient outcomes. Models like J48 with ensemble boosting (99.16%) and MLP ensemble bagging (98.6%) exceed their individual classifier equivalents, demonstrating that the integration of many learners enhances generalization and precision. In contrast, simpler models like KNN with PCA (85.71%), SVM, and Decision Trees show average performance.

Evaluation Result of Machine Learning Models for Dual Prediction Tasks

The analyzed research indicates that ensemble-based tree models and deep learning architectures are the most efficient methods for predicting both employee performance and turnover, especially when integrated with data balance, feature optimization, or attention processes.

A wide variety of high-performing approaches is documented in relation to employee turnover. Deep learning methodologies, such as Deep Neural Networks (DNN) and WTTE-RNN, provide enhanced predictive efficacy, especially in modeling complex and temporally dependent employee behavior. Moreover, hybrid and optimized models, like Genetic Algorithm LightGBM (GA-LightGBM), hybrid ensemble frameworks, and enhanced decision tree variations (Improved CART-DT), exhibit significant accuracy enhancements compared to traditional techniques. Ensemble tree-based techniques, particularly Random Forest (RF) and Gradient Boosting models (e.g., XGBoost), have been identified as superior performers. Although XGBoost frequently achieves the highest predictive accuracy, several studies emphasize the importance of explainability-oriented approaches, such as SHAP, that can enhance Logistic Regression and Random Forest models. These approaches are valued not only for their predictive performance but also for their ability to provide transparent and interpretable insights.

The superior models for predicting employee performance mostly use ensemble and deep learning approaches. Gradient Boosting Machines (GBM) and Random Forest models, when integrated with data balancing methods (RF-ROS), regularly achieve increased prediction accuracy. Deep learning architectures, such as Artificial Neural Networks (ANN), Deep Neural Networks (DNN), and RNN-LSTM models, demonstrate robust performance, particularly in research using longitudinal or sequential performance data. Furthermore, research using feature reduction and ensemble learning techniques, such as PCA with KNN and J48 paired with AdaBoost, indicates significant performance improvements, emphasizing the effectiveness of hybrid modeling approaches.

CONCLUSION

This systematic literature review reveals the extensive and growing application of machine learning in predicting employee performance and attrition, marked by a methodological shift from traditional algorithms like Logistic Regression, Decision Trees, Support Vector Machines, K-Nearest Neighbors, and Naive Bayes—valued for interpretability—to superior tree-based ensemble models and deep learning techniques that deliver reliable performance across both tasks. Distinct yet overlapping input features emerge, with turnover prediction prioritizing longitudinal, behavioral, and organizational factors, performance prediction focusing on productivity, task-related, and evaluative indicators, and both domains relying on classification metrics such as accuracy, precision, recall, F1-score, and AUC-ROC. The analysis highlights a trend toward hybrid, ensemble, and explainable ML models in HR analytics, enabling integrated frameworks that simultaneously evaluate performance and attrition risk to bolster proactive decision-making and strategic workforce management. For future research, scholars should investigate causal inference techniques, such as propensity score matching or instrumental variable analysis integrated with ML, to move beyond correlational predictions and uncover actionable mechanisms driving performance-turnover dynamics in diverse organizational contexts.

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